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Encrypted speech authentication and recovery scheme based on fragile watermarking

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Abstract
In order to solve the problems that most of the existing speech authentication algorithms can only realize tamper localization of frames, and need to the additional recovery information during tamper recovery, an encrypted speech authentication and recovery scheme based on fragile watermarking was proposed by using least significant bit method (LSB) and linear interpolation method. Firstly, a scrambling encryption algorithm based on Henon mapping is designed to encrypt and decrypt speech. Secondly, the frame number of ciphertext speech is used to construct fragile watermarking information. Finally, the LSB method is used to embed the watermarking into the fourth place after the decimal point of encrypted speech for authentication, and residial-based linear interpolation method is used to tamper recover the unauthenticated speech. The experimental results show that the proposed scheme can realize the tamper detection and location of sampling points, and the restored tampered content has good auditory quality.

Keywords Encrypted speech authentication · Fragile watermarking · Least significant bit · Tamper recovery · Linear interpolation

1 Introduction

With the growing maturity of multimedia technology and the development of cloud technology, more and more people choose to store multimedia files on the cloud platform. However, the security problems faced by the cloud platform have seriously restricted its development, and the speech uploaded to the cloud is extremely vulnerable to illegal attack and malicious tampering. Audio tampering can be roughly divided into two categories: carrier-based and content-based [1]. Carrier-based tampering refers to tampering with the structure of the audio files and related description of carrier speech. Content-based tampering refers to the content modification of carrier speech, such as mute, substitution and deletion. For content-based speech tampering, the synchronization verification information and tamper recovery information are embedded into the carrier speech as fragile watermarking, which can realize tamper detection and tamper recovery and effectively protect the speech files.

In recent years, fragile watermarking has made numerous research results in speech authentication. Shi et al. [2] used perceptual hashing and robust watermarking to perform speech authentication in plaintext domain, the watermarking method has good robustness, and it can locate the tamper accurately. Qian et al. [3] proposed a digital watermarking based on integer wavelet transform in encrypted domain, which embeds the expected value between encrypted speech frames as a digital watermarking into the encrypted domain, this method has strong ability to locate the tampered area without using sync code. Hu et al. [4] proposed a multipurpose blind watermarking algorithm based on lifting wavelet transform, the algorithm uses three watermarking algorithms for watermark embedding, in which the fragile watermarking is embedded into the third and second level detail subbands using adaptive quantization index modulation, and the synchronization code is embedded into the third level approximation subband through rational dither modulation. Meng et al. [5] aimed at speech splicing attack, compared the variance of background noise to detect speech tampering, this method can effectively locate speech splicing tampering. Wang et al. [6] proposed a spectral manipulations

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based information hiding method for tampering detection in sparse domain, the original speech is decomposed into sparse component and low-rank component by the robust principal component analysis, and the watermark information is embedded in the sparse component, which can detect tamper through spectral operation. Wang et al. [7] proposed an automatic classification framework of support vector machine (SVM) based on feature fusion, which fused the phase and instantaneous frequency features of the electric network frequency component and classified them by SVM classifier to realize audio tamper detection, however, the accuracy of this method is poor for low SNR signals. Li et al. [8] compared the difference between the individual syllables to determine whether the audio was copy-move forgery. Yan et al. [9] extracted pitch sequence and formant sequences as the feature set, and used dynamic time warping to detect and locate copy-move forgery in speech. Esmaeilbeig et al. [10] proposed a tamper detection algorithm based on singular spectrum analysis, the fragile watermarking is embedded by modifying the relatively unimportant singular value of the original speech signal, which has good inaudibility but poor robustness. Liu et al. [11] mapped the frame number as integer sequence and embedded it into different segments of each frame for tamper detection. Wang et al. [12] frames the speech and divided into segments, scrambled each segments of speech, calculated the self-correlation of the scrambled signal, and embedded fragile watermarking by quantizing the self-correlation, this method has good inaudibility. Wang et al. [13] proposed a speech compression algorithm based on discrete wavelet transform (DWT), which realized speech compression and built recovery signal by DWT, realized tamper detection and recovery by using frame number and compressed signal. Qian et al. [14] proposed an encrypted speech authentication method, the watermarking of this method is generated by encrypted speech features and embedded in the detail coefficients of integer wavelet transform, which can effectively locate the tamper, and the tamper can be recovered by the correlation between the sampling points without additional information. Menendez-Ortiz et al. [15] used the encoded information as fragile watermarking for authenticating the contents and embedded it in the integer discrete cosine transform (IDCT) domain, but the recovery effect is poor when the tampered area is large. Menendez-Ortiz et al. [16] used auditory masking properties to select frequency and map it to the IDCT domain for fragile watermark embedding, auditory masking properties can determine the frequency of embedding watermark better. Brajović et al. [17] used the gradient based on the DCT sparsity measure to detect tampering, and reconstructed the tampered location through compressive sensing. Lu et al. [18] embedded the DCT coefficients of the original speech as watermark information for tampering recovery by LSB method, extracted the watermark information of tampered area by compressive sensing technology and sparseness in the DCT domain of tampered area, but the recovered speech signal has noise. Liu et al. [19, 20] proposed a residue restoration method based on DWT, which takes the approximate coefficient of DWT as the compressed signal, recovers the compressed signal and takes the residual with the original signal, respectively takes the approximate coefficient of residual DWT as the compressed signal and the residual are subjected to DCT as the compressed signal, and forms the recovered signal with DWT compressed signal, embeds the watermarking by LSB method. Qian et al. [21] generated the reference information for tamper recovery by compressing the speech sample points, and embedded it into the LSB of encrypted speech. Fan et al. [22] used set partitioning in hierarchical trees encoding and repeated coding to generate compressed signal for speech recovery, used hash algorithm and chaotic sequence to generate check bits for synchronization counterfeiting attack. Gomez-ricardez et al. [23] proposed a self-recovery watermarking scheme for the discordant size content replacement attacks, which generates compressed recovery data by source-channel coding, and synchronizes the watermark by synchronization strategy in tampering recovery stage, but this method must know the length of the original speech. Ma et al. [24] used time-varying vocal vibration signal as sound source excitation and the fitted formant frequency to synthesize and recover speech, which has good applicability in tamper recovery. Firouzeh et al. [25] used the maximum feasible subsystem problem to recover the sparse compressed speech, this method can recover the compressed signal with high quality by fewer measured signals and this method has good applicability in tamper recovery.

Based on the above literature analysis, the existing speech tampering detection algorithms can achieve the accurate tamper location of the speech frames, but most tamper location algorithms based on fragile watermarks are difficult to achieve due to the need for certain sampling points to embed tamper recovery information. However, the tampering localization algorithm that can realize the tampering of sampling points is often time-consuming, and it's difficult to achieve efficient and accurate tamper location of sampling points. Most of the existing speech tampering recovery schemes require additional recovery information to recover, and the additional information is often embedded into the speech together with the information used for tamper location as a fragile watermark. The watermark embedding capacity and transparency in the speech interact, while more watermarks are embedded, the transparency of the watermark is reduced; at the same time, in the limited embedding capacity, the embedding capacity of the information used for tampering detection and the information used for tampering recovery is difficult to balance. If a certain location and the location containing the location recovery information are tampered together, tamper recovery cannot be performed. Therefore, it

should be considered not to use extra information for tamper recovery, leaving a large amount of embedding capacity to embed tamper detection information.

To solve the above problems, we presents an encrypted speech authentication and recovery scheme based on fragile watermarking by LSB and linear interpolation. The proposed scheme uses the scrambling encryption algorithm based on Henon mapping, which effectively reduces the correlation between the sampling points on the premise of ensuring the security of speech and watermark, uses LSB to embed the fragile watermarking, uses linear interpolation method based on residual to recover the encrypted speech after tampering detection accurately through the correlation between the sampling points. Experimental results show that the proposed scheme can achieve the tampering recovery of speech without additional information, while ensuring high-quality recovery of the tampered speech, it does not require any tampering recovery information, and leaves more embedding space to embed fragile watermarks as tampering detection information. Accurate tampering detection of sampling points is achieved, it has excellent transparency, and the watermark information is difficult to detect.

The remaining part of this paper is organized as follows. Section 2 describes the relevant theories in detail, including Henon mapping, LSB and linear interpolation. The detailed encrypted speech authentication and recovery scheme is described in Sect. 3. Section 4 gives the experimental results and the performance analysis compared with other related methods. Finally, we conclude our paper in Sect. 5.

2 Related theories

2.1 Henon mapping

The chaotic system refers to the random and irregular movement in a certain system, which is usually used in encryption methods. Henon mapping [26] is a kind of two-dimensional chaotic map, which is mainly used in image encryption, the calculation formula is shown in Eq. (1).

$$\begin{cases} x_{n+1} = 1 - ax_n^2 + y_n \\ y_{n+1} = bx_n \end{cases} \tag{1}$$

where the parameters a and b are in the range of $1.54 < a < 2$, $0 < |b| < 1$, the system is in the typical state of Henon chaotic map at this time, the system is in the hyperchaotic state, and the chaotic sequence generated by the initial values x_0 and y_0 has strong randomness and sensitivity to the initial values.

Compared with other chaotic maps, the Henon mapping is easy to implement, has better security and higher system complexity, and is suitable for generating scrambling sequences of speech encryption.

2.2 Least significant bit method

LSB [13] can embed certain information in the carrier multimedia files and ensure certain transparency at the same time, which embeds the information by replacing the LSB of the target multimedia files. When extracting the embedded information, which can directly extract from the carrier files without other auxiliary information. The embedding and extraction of the watermark by this method is simple and efficient, and the embedded watermark has a certain degree of inaudibility, but the robustness is extremely poor, and only suitable for embedding fragile watermarking, When undergo all kinds of attacks, the watermark information extracted have great changes, which can effectively determine the various attack methods of the carrier's speech. In this paper, the carrier speech is embedded with watermark in the form of encrypted, and the encrypted speech cannot be used for various speech content preserving operations, the LSB-based watermark embedding method doesn't need to consider the robustness, which effectively avoid the disadvantage of poor robustness of LSB.

For the LSB method of speech, if the quantization bit of the target speech is 16 bit, each sampling point has 16 bits, the LSB method modifies the 16th bit of each sampling point while keeping the first 15 bits unchanged, then the LSB embedding of speech can be realized.

2.3 Linear interpolation

Linear interpolation [14] is the method of obtaining the target point data approximately through the data of the two points before and after the target point. If two points (x_1, y_1) and (x_3, y_3) are known, the formula for calculating the approximate value y_2 of a point x_2 in the interval $[x_1, x_3]$ is shown in Eq. (2).

$$y_2 = y_1 + \frac{(x_2 - x_1)y_3 - (x_2 - x_1)y_1}{x_3 - x_1} \tag{2}$$

Due to this paper only through the correlation between the sampling points, the additional information isn't needed to recover the tampered speech, and compared with other methods of approximation and fitting, linear interpolation can efficiently recover the tampered sampling points according to the adjacent sampling points, this paper chooses the linear interpolation method to recover the tampered speech sampling points, and the sampling points recovered by this method are closer to the true values.

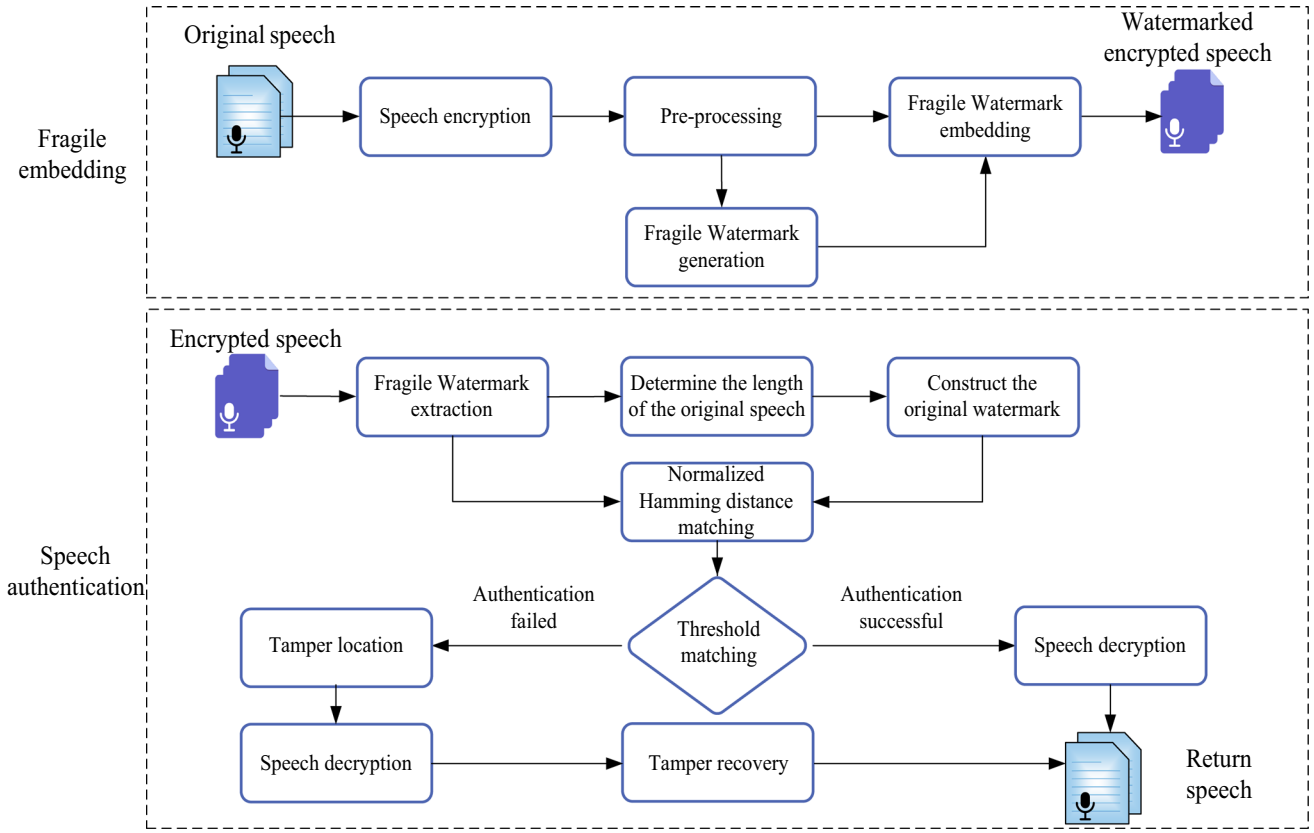


Fig. 1 The processing flow chart of encrypted speech authentication scheme

3 The proposed scheme

Figure 1 shows the processing flow chart of encrypted speech authentication scheme. This scheme combines the existing encrypted speech authentication model, uses the frame length of original speech as fragile watermarking for authentication, and uses linear interpolation method based on residual to tamper recover the speech that failed authentication. The algorithm process is mainly composed of four parts: speech encryption and decryption, fragile watermark generation and embedding, encrypted speech authentication and tamper recovery.

The processing flow of encrypted speech authentication scheme is as follows:

Step 1: Speech encryption. Encrypt the original speech.
Step 2: Fragile watermark generation and embedding. The encrypted speech is preprocessed, and the fragile watermarking is constructed by the frame number and embedded in encrypted speech to construct watermarked encrypted speech.
Step 3: Encrypted speech authentication. Extract the fragile watermarking of the encrypted speech to be authenticated, determine the length of the original speech through the extracted fragile watermarking, then construct the original watermarking based on the frame length of original speech, and calculate the normalized Hamming distance between the extracted fragile watermarking and the original watermarking to realize authentication, and the authenticated encrypted speech is decrypted and returned to the user.
Step 4: Tamper recovery. The tamper location is performed on the encrypted speech that has failed the tamper detection, then the tampered encrypted speech is decrypted and the tampered sampling point is recovered, the decrypted and recovered speech is returned to the user.

3.1 The scrambling encryption algorithm based on Henon mapping

Since the principle of the tamper recovery scheme in this paper is to perform tamper recovery through the correlation between the sampling points after decrypting the speech, the encryption algorithm needs to reduce the correlation between the encrypted speech sampling points as much as possible. In this paper, twice scrambling encryption is performed through Henon mapping, which can effectively eliminate the correlation between original speech sampling points.

3.1.1 Speech encryption

Step 1: Speech pre-processing. The speech signal S is divided into frames without overlapping, in which the length of speech signal is L and the frame length is 128, the speech SS of $L/128 \times 128$ is generated.
Step 2: Scrambling encryption. Select the initial values of key $[x_0, y_0]$ and the parameters a_1, b_1 of the Henon sequence, use the chaotic map of Sect. 2.1 to generate the first dimensional chaotic sequence $X = \{x_1, x_2, x_3, \ldots, x_{128}\}$ and the second dimensional chaotic sequence $Y = \{y_1, y_2, y_3, \ldots, y_{L/128}\}$, sort two chaotic sequences in ascending order to obtain new sequences XS and YS, the original position index I_x and I_y corresponding to each element of the sequence XS and YS are used as scrambling sequences, the inter-frame scrambling is performed through I_x, and the intra-frame scrambling is performed through I_y, the speech SS' after encryption is obtained.
Step 3: Secondary scrambling encryption. Transpose the encrypted speech SS', the dimension of SS'^{T} after transposition is $128 \times L/128$, select the initial values of the key $[u_0, v_0]$ and the parameters a_2, b_2 of the Henon sequence, use the chaotic map of Sect. 2.1 to generate the first dimensional chaotic sequence $U = \{u_1, u_2, u_3, \ldots, u_{L/128}\}$ and the second dimensional chaotic sequence $V = \{v_1, v_2, v_3, \ldots, v_{128}\}$, sort two chaotic sequences in ascending order to obtain new sequences US and VS, the original position index I_u and I_v corresponding to each element of the sequence US and VS are used as scrambling sequences, the inter-frame scrambling is performed through I_u, and the intra-frame scrambling is performed through I_v, the speech SS'' after encryption is obtained.
Step 4: Reconstruct the speech. Convert the $128 \times L/128$ encrypted speech SS'' into the $L \times 1$ matrix and reconstruct the encrypted speech E.

3.1.2 Speech decryption

Step 1: Encrypted speech pre-processing. The encrypted speech signal E is divided into frames without overlapping, in which the length of speech signal is L and the frame length is 2500, the speech EM of $L/2500 \times 2500$ is generated.
Step 2: Scrambling decryption. Select the initial values of key $[u'_0, v'_0]$ and the parameters a_2, b_2 of the Henon sequence, use the chaotic map of Sect. 2.1 to generate the first dimensional chaotic sequence $U' = \{u'_1, u'_2, u'_3, \ldots, u'_{2500}\}$ and the second dimensional chaotic sequence $V' = \{v'_1, v'_2, v'_3, \ldots, v'_{L/2500}\}$, sort two chaotic sequences in ascending order to obtain new sequences US' and VS', the original position index I'_u and I'_v corresponding to each element of the sequence US' and VS' are used as scrambling sequences, the inter-frame scrambling is performed through I'_u, and the intra-frame scrambling is performed through I'_v, the speech EM' after decryption is obtained.
Step 3: Secondary scrambling decryption. Transpose the encrypted speech EM', the dimension of EM'^{T} after transposition is $2500 \times L/2500$, select the initial values of the key $[x'_0, y'_0]$ and the parameters a_1, b_1 of the Henon sequence, use the chaotic map of Sect. 2.1 to generate the first dimensional chaotic sequence $X' = \{x'_1, x'_2, x'_3, \ldots, x'_{L/2500}\}$ and the second dimensional chaotic sequence $Y' = \{y'_1, y'_2, y'_3, \ldots, y'_{2500}\}$, sort two chaotic sequences in ascending order to obtain new sequences XS' and YS', the original position index I'_x and I'_y corresponding to each element of the sequence XS' and YS' are used as scrambling sequences, the inter frame scrambling is performed through I'_x, and the intra-frame scrambling is performed through I'_y, the speech EM'' after decryption is obtained.
Step 4: Reconstruct the speech. Convert the $2500 \times L/2500$ decrypted speech EM'' into the $L \times 1$ matrix and reconstruct the speech S.

3.2 Fragile watermarking algorithm based on the LSB

In this paper, the sampling rate of the watermark carrier speech is 16 bit, and the speech quality is the least affected after modifying the fourth decimal place, this paper chooses the LSB method to embed the fragile watermarking on the fourth decimal place of the original speech. In order to prevent the least significant bit from being attacked and destroying the watermark information, this paper sets the fifth decimal place to zero when embedding the watermark, and then uses the fourth and fifth decimal place to jointly restore the watermark information.

The tamper recovery method in this paper doesn't need additional information for tamper recovery, only needs to consider the construction and embedding of fragile watermarking for tamper detection, so only embeds the frame number as fragile watermarking in this paper.

3.2.1 Watermark generation and embedding

Step 1: Pre-processing of encrypted speech. The speech of length L is encrypted according to the method of Step 1, Step 2 and Step 3 in Sect. 3.1.1, to obtain the encrypted matrix SS'' of $L/128 \times 128$.
Step 2: Fragile watermark generation. The frame number matrix w_f is flipped up and down to form the matrix w_{ff}, the matrix w_f and matrix w_{ff} are repeatedly spliced left and right to obtain the watermark w_o of $L/128 \times 8$, $w_o = [w_f, w_{ff}, w_f, w_{ff}, w_f, w_{ff}, w_f, w_{ff}]$, and then reconstruct the watermark

into the matrix w_b of $L/128 \times 32$ according to the thousands, hundreds, tens, and ones bits of each data in the watermark matrix w_o, of which less than four digits are set the insufficient digits to 0, and each data in the matrix w_b is converted into a four bit binary matrix to form the watermark information w of $L/128 \times 128$.

Step 3: Watermark embedding. The watermark embedding is realized by modifying the fourth decimal place f of each sample value of the encrypted matrix SS'' by the watermark w bit by bit, the calculation formula of watermark embedding is shown in Eq. (3), f' is the fourth decimal place of the embedded watermark, and then the fifth decimal place of the embedded encrypted matrix sampling point is set to 0 to obtain the watermarked encrypted speech matrix SSW''.

$$f' = \begin{cases} f, & f \text{ is even} \\ f+1, & f \text{ is odd} \end{cases} w = 0;$$
$$f' = \begin{cases} f, & f \text{ is even} \\ f+1, & f \text{ is odd} \end{cases} w = 1 \quad (3)$$

Step 4: Reconstruct the encrypted speech with watermark. The watermarked encrypted speech matrix SSW'' is processed according to Step 4 of Sect. 3.1.1 to obtain the watermarked encrypted speech EW.

3.2.2 Watermark extraction

Step 1: Pre-processing of encrypted speech. The watermarked encrypted speech EW is divided into frames according to the method of Step 1 in Sect. 3.1.1, and the encrypted speech matrix EWM of $L/128 \times 128$ is obtained.

Step 2: Watermark extraction. Extract the fourth decimal place f'_1 and the fifth decimal place f'_2 of the watermarked encrypted speech matrix EWM to construct the watermarked data f'', the construction formula is shown in Eq. (4), and then according to the watermark extraction Eq. (5) extract the watermark w' through the watermarked data f''.

$$f'' = \begin{cases} f'_1 + 1, & (0.1 \times f'_2,) > 0.5 \\ f'_1, & (0.1 \times f'_2,) < 0.5 \end{cases} \quad (4)$$

$$\begin{cases} w' = 0, & f'' \text{ is even} \\ w' = 1, & f'' \text{ is odd} \end{cases} \quad (5)$$

3.3 The authentication of encrypted speech

The proposed authentication scheme only needs to extract the fragile watermarking in the encrypted speech to be authenticated, and the length of the original speech can be judged by the watermark, to construct the original watermark information for tamper detection. Compared with other tamper detection algorithms that require additional information or need to know the length of the original speech, the method in this paper enhances the security of the speech. At the same time, since the tamper recovery algorithm in this paper also doesn't require additional information, there is a larger space for embedded synchronization information in the speech. Compared with other tamper localization algorithms, this method realizes tamper location of sampling points step by step, which has higher tamper accuracy, which provides guarantee for high quality tamper recovery.

3.3.1 Tampering detection

Step 1: Pre-processing of watermarked encrypted speech. Mirror flip the watermarked encrypted speech EW of length L' to obtain the encrypted speech EW', the encrypted speech EW and EW' are divided into frames according to the method of Step 1 in Sect. 3.1.1, and divided into LL frames after discarding the data of less than one frame, the $LL \times 128$ watermarked encrypted speech matrix EWM and EWM' are obtained, the matrix EWM' is flipped left and right to obtain the watermarked encrypted speech matrix EWM''.

Step 2: Watermark extraction. The watermarked encrypted speech matrix EWM and EWM' are extracted according to the method of Step 2 in Sect. 3.2.2, and every four binary watermark data of each frame of the extracted watermarks w'_1 and w'_2 are reconstructed into $LL \times 32$ decimal watermark data w'_{b1} and w'_{b2}, every four decimal watermark data is reconstructed into $LL \times 8$ frame number watermark data w'_{o1} and w'_{o2} according to thousands, hundreds, tens, and ones bits.

Step 3: Determine the length of original speech. The data of each dimension in the frame number watermarking data $\boldsymbol{w'}_{o1}$ and $\boldsymbol{w'}_{o2}$ are averaged to obtain two speech length matrices of $LL \times 1$, and determine the data c with the highest frequency in the speech length matrix, then the calculation formula of the length L_o of original speech is shown in Eq. (6).

$$L_o = (c-1) \times 2500 \quad (6)$$

Step 4: Judgment of tamper attack. Comparing the length of the encrypted speech L' with the original speech length L_o, if $L' > L_o$, the encrypted speech has been insertion attacked, and the number of inserted sampling points $\Delta = L'—L_o$; if $L' < L_o$, the encrypted speech has been deletion attack, the number of deleted sampling points $\Delta = L_o—L'$; if $L' = L_o$, the encrypted speech has been mute attacked, substitution attacked or not attacked.

3.3.2 Tampering location

Step 1: Reconstruct the original watermarking. Construct the watermarking w_o, w_b and w according to the method of Step 2 in Sect. 3.2.1 of the length L_o of original speech.
Step 2: Locate the tampered frames. The constructed watermark w_o is compared with the extracted watermark w'_{o1}, when the speech is attacked by insertion or deletion, the starting frame f_f of the tampered position is searched, and the fuzzy position of the tampered starting point is $[f_f, d_1]$. If $L' = L_o$ and the two watermarks are exactly the same, the speech isn't attacked; when the two watermarks are different, find the start frame f_f and the end frame f_d of the tampered position, and the fuzzy position of the starting and ending points are (f_f, d_1) and (f_d, d_2).
Step 3: Locate the sampling point where the insertion attack or deletion attack has been tampered with. If the speech is attacked by insertion or deletion, by comparing the data in the f_f frame $[4 \times d_1–3{,}4 \times d_1]$ of the watermark w_b and w'_{b1}, the tamper starting point position d'_1 is further determined, and finally by comparing the data in the f_f frame $[4 \times d'_1–3{,}4 \times d'_1]$ of the watermark w and w'_1, the tamper starting sampling point d''_1 is determined, then the tamper starting point position is x_{zf}, and the calculation formula is shown in Eq. (7), the tampered area is $[x_{zf}, x_{zf} + \Delta]$.

$$x_{zf} = 128 \times f_f + d''_1 - 128 \tag{7}$$

Step 4: Locate the sampling point where the mute attack or substitution attack has been tampered with. If the speech is attacked by mute or deletion, follow the method in Step 3 to determine the sampling point d''_1 of the tamper starting, then use the fuzzy position of the end (f_d, d_2) to determine the sampling point d''_2 of the tamper ending according to the method in Step 3, determine the tampered start and end positions x_{zf} and x_{zd}, the tampered area is $[x_{zf}, x_{zd}]$.

3.4 Tampered recovery

The linear interpolation method based on residual recovery is used to recover the tampered speech in this paper, the tamper recovery through the correlation between the two sampling points around the tampered sampling point. In order to improve the accuracy of the tampering, the residual method is used in this paper to optimize the results of linear interpolation, and the recovery effect of the tampered sampling points is better through the correlation of the surrounding four sampling points.

Step 1: Pre-processing of tamper recovery. Set the tampered sampling point of the encrypted speech *EW* after tamper detection to infinity, and perform speech decryption according to the method of Sect. 3.1.2, diffused the tampered sampling points into the whole speech.
Step 2: Find the fitted value through linear interpolation. The fitted value y_z of the tampered sampling point x_z is calculated by the sampling points x_{z-} and x_{z+1} that aren't infinite on both sides of the tampered sampling point x_z in speech (skip the point when a sampling point is infinite), and the calculation formula of the fitted value y_z is shown in Eq. (8).

$$y_z = \frac{x_{z-1} + x_{z+1}}{2} \tag{8}$$

Step 3: Residual recovery. The fitted values y_{z-1} and y_{z+1} of the tampered sampling point x_z are calculated by the sampling points $x_{z-2}, x_{z+1}, x_{z-1}$ and x_{z+2} that aren't infinite on both sides of the tampered sampling point x_z in speech (skip the point when a sampling point is infinite), then the fitted value calculation formula of tampered sampling points is shown in Eq. (9).

$$\begin{aligned} y'_z &= y_z + \frac{x_{z-1} - y_{z-1} + x_{z+1} - y_{z+1}}{2} \\ &= \frac{3 \times x_{z-1} + 3 \times x_{z+1} - x_{z-2} - x_{z+2}}{4} \end{aligned} \tag{9}$$

Step 4: Reconstruct the speech. The above operations are performed on all the tampered sampling points in the tampered speech, and the sampled points after tamper recovery are reconstructed the speech by D/A conversion.

4 Experimental results and performance analysis

In the experiment, the speech library is constructed from the speech in the THCHS-30 speech library [27]. This paper use a single-channel WAV format speech segment with the frequency of 16 kHz, and the sampling accuracy of 16 bit, each speech is 20 s. In order to fully demonstrate the performance of the proposed scheme, 1000 speech are randomly selected as a database for the experiments of inaudibility, quality of tamper recovery, and security, and one 20 s speech is randomly selected for the experiments of tamper location. Experimental hardware environment: Intel(R) Core(TM) i5-7300HQ CPU, 2.50 GHz, 8 GB of memory. The software environment is: Windows 10, MATLAB R2017a.

4.1 Payload capacity

The payload capacity is one of the important indicators to measure the audio watermarking algorithm, the payload capacity *Cap* of the method in this paper is related to the

Table 1 Comparison of the inaudibility with other methods

Methods	SNR/dB	PESQ
Ref. [13]	29.036	4.216
Ref. [14]	52.46	4.483
Ref. [18]	41.54	4.391
Ref. [19]	28.8783	4.232
Ref. [20]	28.9909	4.238
Ref. [21]	47.43	4.463
Ref. [30]	–	4.500
Proposed	62.5671	4.500

time of the carrier speech t, the number of check bits per frame v, the number of carrier speech frames N and other parameters. The calculation formula of the payload capacity of this paper is shown in Eq. (10), $t = 20$, $v = 128$, $n = 2500$, $Cap = 16{,}000$, each sampling point is embedded with 1 bit watermark information.

$$Cap = \frac{v \times N}{t} \tag{10}$$

4.2 Inaudibility

Inaudibility mainly refers to the degree to which the watermarked speech auditory quality decreases after the watermark is embedded, this paper selects the signal-to-noise ratio (SNR) and perceptual evaluation of speech quality (PESQ) which can directly and accurately reflect the change of watermarked speech quality to evaluate the watermarked speech. PESQ is an objective MOS value evaluation method with the value from -1 to 4.5. When PESQ > 3.5, the speech quality of watermarked speech is better. Table 1 shows the inaudibility of the proposed scheme with the other method in Refs. [13, 14, 18–21, 30]. The calculation formula of SNR is shown in Eq. (12), and L is the length of the original speech, l is the sampling point, x is the original speech, and y is the watermarked speech.

$$SNR = 10\log_{10}\frac{\sum_{l=1}^{L} x^2(l)}{\sum_{l=1}^{L}\left(x^2(l) - y(l)\right)^2} \tag{12}$$

It can be seen from Table 1 that the SNR and PESQ of the proposed method are higher than Refs. [13, 14, 18–21], same as Ref. [30] which based on perceptual hashing, the proposed scheme has good transparency. This is because the proposed scheme only needs to embed binary synchronization information, while other methods need to embed compress recovery information, the proposed scheme only needs to modify the fourth and fifth digits after the speech sampling point, and doesn't need to modify other data. Therefore, the proposed scheme is highly inaudibility, and the watermarked speech has basically unchanged in the auditory quality from the original speech.

4.3 Tamper location analysis

Compared with other methods, tamper detection in the proposed scheme doesn't need to know the length of original speech or any additional information, only detect the extracted fragile watermarking, the length of original speech can be judged and the fragile watermarking for authentication is constructed.

When the speech is failed authentication and tampered is detected, the speech needs to locate the tamper, and then the higher quality tamper recovery can be performed after the accurately tamper location. In order to prove the tamper location ability of the proposed scheme, MATLAB R2017a is used in the experiment to tamper with the watermarked encrypted speech. The tamper with the way mainly include mute attack, substitution attack, insertion attack, and deletion attack, the attack range is 7.5% of the original speech sampling point.

1. Mute attack

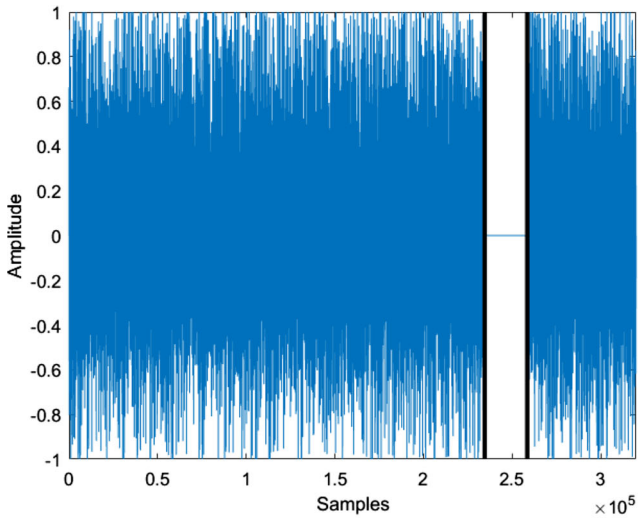
The mute attack represents that some sampling points in the watermarked speech are set to zero. In this paper, 24,000 sampling points within the range of the watermarked encrypted speech [234,561, 258,561] are muted. Figure 2 shows the speech waveform and tamper location after the speech is muted.

As shown in Fig. 2, the tamper detection algorithm in this paper detects the sampling points within the range of the encrypted speech [234,565, 258,561] are muted, there is an error of four sampling points from the starting point of the actual attacked position.

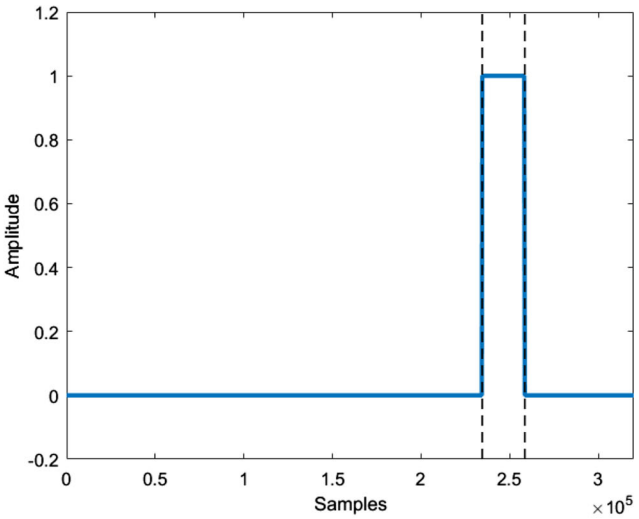
2. Substitution attack

The substitution attack represents that some sampling points in the watermarked speech are replaced by random number or other sampling points of speech signals. In this paper, 24,000 sampling points within the range of the watermarked encrypted speech [123,456, 147,456] are replaced. Figure 3 shows the speech waveform and tamper location after the speech is replaced.

As shown in Fig. 3, the tamper detection algorithm in this paper detects the sampling points within the range of the encrypted speech [123,458, 147,457] are replaced, there is an error of two sampling points from the starting point of the actual attacked position and an error of one sampling points from the ending point of the actual attacked position.

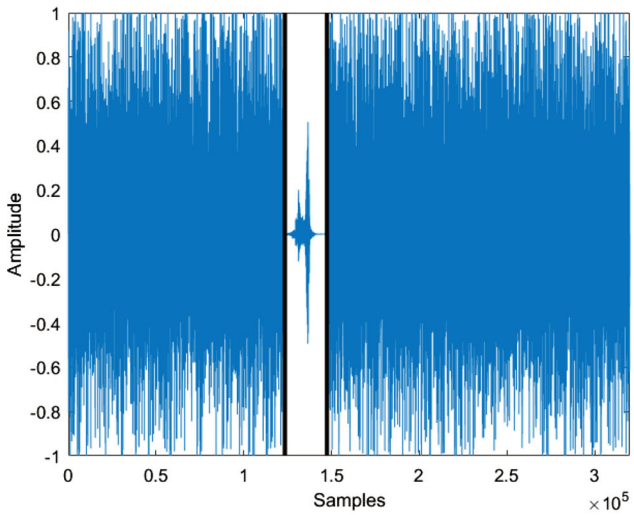


(a) The speech waveform of mute attack

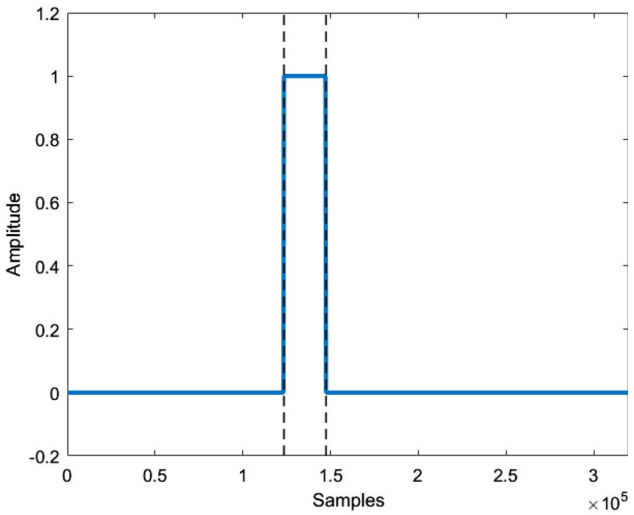


(b) The location of tampered speech frame

Fig. 2 The speech waveform and tamper location after the mute attack



(a) The speech waveform of substitution attack



(b) The location of tampered speech frame

Fig. 3 The speech waveform and tamper location after the substitution attack

3. Insertion attack

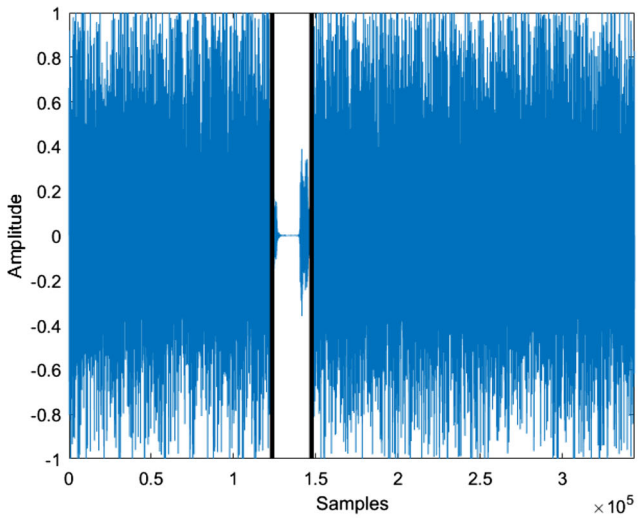
The insertion attack represents that inserted by inserting sampling points of other speech signals at the certain position of the watermarked speech. This paper inserts 24,000 sampling points after the 123,456 sampling point of the watermarked encrypted speech. Figure 4 shows the speech waveform and tamper location after the speech is inserted.

As shown in Fig. 4, the tamper detection algorithm in this paper detects that the encrypted speech has been attacked, and the sampling points within the range of the encrypted speech [123,458, 147,458] are the inserted sampling points, there is an error of two sampling points from the actual attacked position.

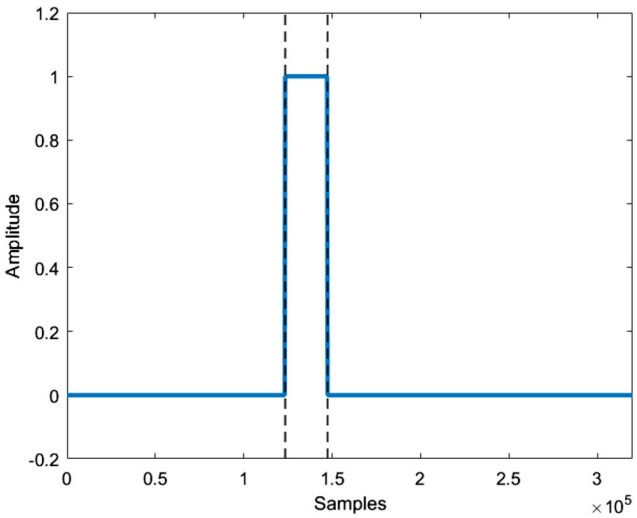
4. Deletion attack

The deletion attack represents that delete the random part of the watermarked speech. This paper deletes 24,000 sampling points within the range of the encrypted speech [234,561, 258,561]. Figure 5 shows the speech waveform and tamper location after the speech is deleted.

As shown in Fig. 5, the tamper detection algorithm in this paper detects that the encrypted speech has been deletion attacked, the sampling point of the deleted sampling point is 234,564 and the number of deleted sampling points is 24,000, there is an error of three sampling points from the actual attacked position.

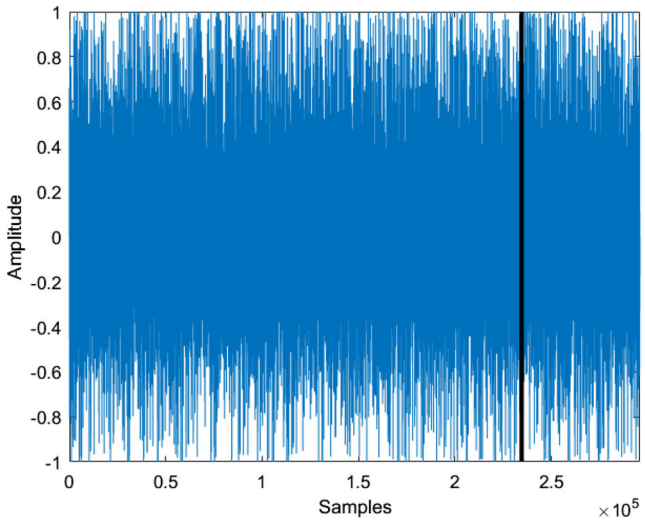


(a) The speech waveform of insertion attack

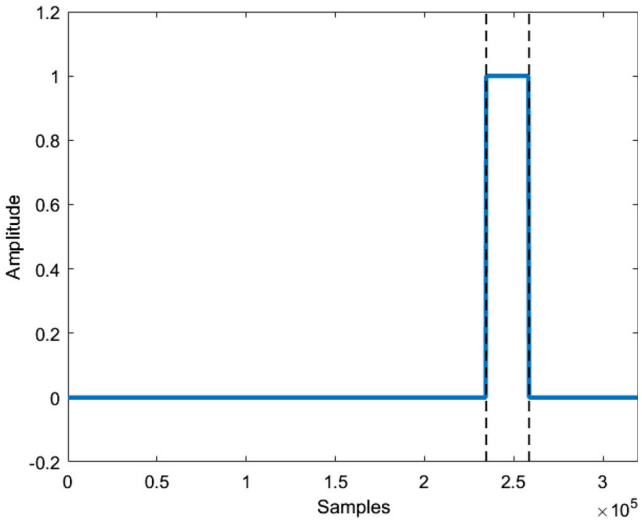


(b) The location of tampered speech frame

Fig. 4 The speech waveform and tamper location after the insertion attack



(a) The speech waveform of deletion attack



(b) The location of tampered speech frame

Fig. 5 The speech waveform and tamper location after the deletion attack

It can be seen from Figs. 2, 3, 4 and 5 that the proposed scheme can realize the tamper detection of mute attack, substitution attack, insertion attack, and deletion attack. Compared with other methods of tamper localization for frames, the proposed scheme has higher tamper localization accuracy, and can locate the tampered sampling point accurately, which lays the foundation for the high-quality tampering recovery.

4.4 Tamper recovery analysis

After the tamper location, the tampered location needs to be restored, in order to better evaluate the recovery quality, this paper selects SNR of the restored signal and the original signal, and PESQ to evaluate speech after tampering recovery. Table 2 shows the compares the SNR under different tampering ratios of the proposed scheme with the other method in Refs. [13, 14, 18–21, 30].

This paper uses MATLAB R2017a to tamper with 1000 test encrypted speech, tampering with the sampling points of 1%, 1.5%, 5%, 7.5%, 10%, and 20% of each speech respectively, the tamper detection and tamper recovery are carried out for the tampered encrypted speech, and recovery quality analysis is carried out for the tampered and restored signals.

As shown in Table 2, the recovery effect of the proposed scheme is better than Refs. [13, 14, 19–21, 30]. when the

Table 2 Comparison of the tamper recovery with existing methods

Tampering ratio	Proposed	Ref. [13]	Ref. [14]	Ref. [18]	Ref. [19]	Ref. [20]	Ref. [21]	Ref. [30]
1%	47.6175	19.7885	38.1541	–	19.8177	19.8147	47.36	39.9526
1.5%	45.8501	18.0059	36.3863	–	18.0480	18.0477	47.37	38.4823
5%	40.6037	12.7715	31.1499	–	12.7678	12.8043	–	32.8832
7.5%	37.3180	11.0152	29.3932	–	11.0296	11.0510	–	30.9430
10%	35.7565	9.7544	28.1403	24.95	9.7767	9.7967	–	29.6660
20%	14.7540	6.7578	14.4529	20.02	6.7860	6.7922	–	14.4018

Table 3 Comparison of PESQ under different tampering ratios

Tampering ratio	1%	1.5%	5%	7.5%	10%	20%
PESQ	4.492	4.486	4.463	4.427	4.431	3.165

Table 4 Performance of encrypted speech and decrypted speech

Methods	PESQ	SNR
Ref. [28]	1.0305/4.5000	− 2.5062/–
Ref. [29]	0.5606/4.4628	− 12.1349/71.5406
Proposed	1.063/4.5000	− 3.0104/–

tampering ratio is 1%. When the tampering ratio is 1.5%, the SNR of the restored signal is better than Refs. [13, 14, 19, 20, 30], and slightly worse than Ref. [21]. When the tampering ratio is 5%, 7.5% and 10%, the SNR of the restored signal is better than Refs. [13, 14, 18–20, 30]. When the tampering ratio is 20%, the SNR of the restored signal is better than Refs. [13, 14, 19, 20, 30], and worse than Ref. [18].

Compared with the tamper recovery method in Refs. [13, 14, 18–21, 30], the proposed scheme doesn't require any additional recovery information, nor worry about the destruction of the carrier speech segments which the recovery information is embedded as watermark, and the correct recovery information cannot be extracted, thus affecting the tampering recovery of speech. The proposed scheme mainly relies on the correlation between the tampered sampling point and the four surrounding sampling points of the decrypted speech to recover the tampered sampling point. When the tampered sampling points increases, the probability of other tampered sampling points around the tampered sampling point will increase, leading to the need of other sampling points with poor correlation for tampering recovery. Therefore, the quality of speech recovery is poor when the tampering percentage is higher.

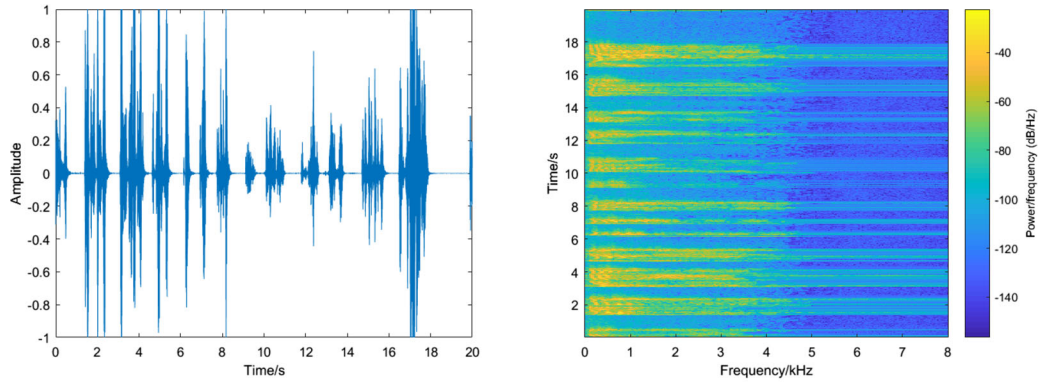
In order to better observe the influence of the increase of tamper percentage on the quality of speech recovery, this paper analyzes the recovery effect of the proposed scheme through the PSEQ of restored speech in Table 3 and the speech waveform and spectrogram in Fig. 6.

As shown in Fig. 6, there is no significant difference in the waveform of speech after tamper recovery, and the pixel points in the spectrogram have little change. It can be seen from Table 3 that when the tampering ratio of the proposed scheme is less than or equal to 10%, the PESQ of the speech after tampering recovery is close to the theoretical value of 4.5, which shows that the proposed scheme has higher tamper recovery performance, and the speech of tamper recovery is close to the original speech. When the tampering ratio is 20%, the performance of tamper recovery is relatively low, but the semantics of the recovered speech can still be accurately recognized.

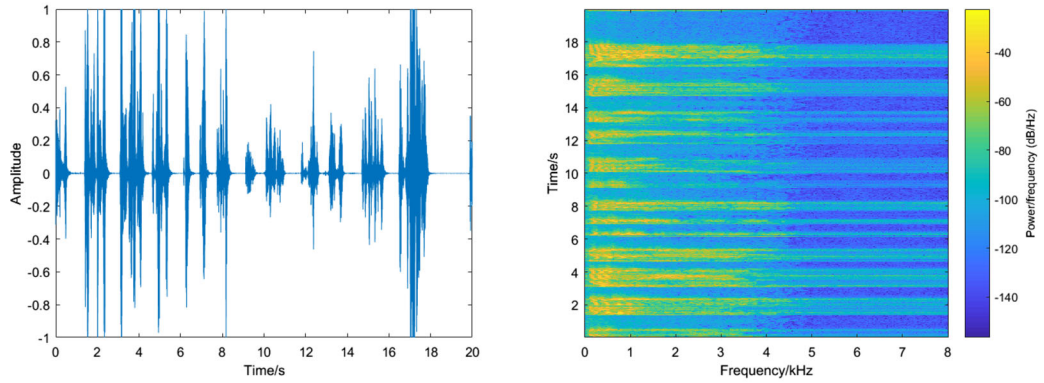
4.5 Security

This paper uses the encryption scheme of Sect. 3.2 to encrypt the original speech, the selected initial values of the key are $x_0 = 0, y_0 = 0, u_0 = 1, v_0 = 1$, the parameters of the Henon sequence are $a_1 = 1.6, b_1 = 0.3, a_2 = 1.6, b_2 = 0.3$, Fig. 7 shows the waveform and spectrogram of original speech and encrypted speech.

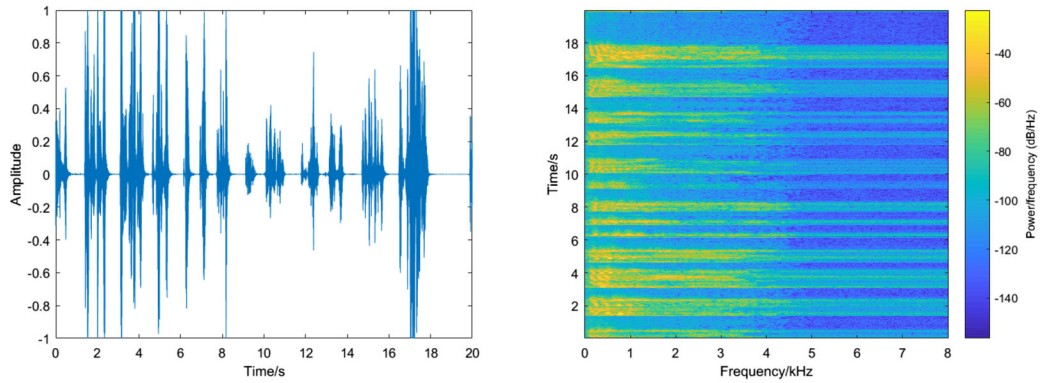
It can be seen from Fig. 7 that compared to the original speech, the encrypted speech waveform is more evenly distributed, which almost destroys the correlation of the original speech sampling points, and the pixel points of the spectrogram are evenly distributed, and any information of the original speech can't be obtained from encrypted speech, proves that the proposed scheme has better encryption effect and higher security.



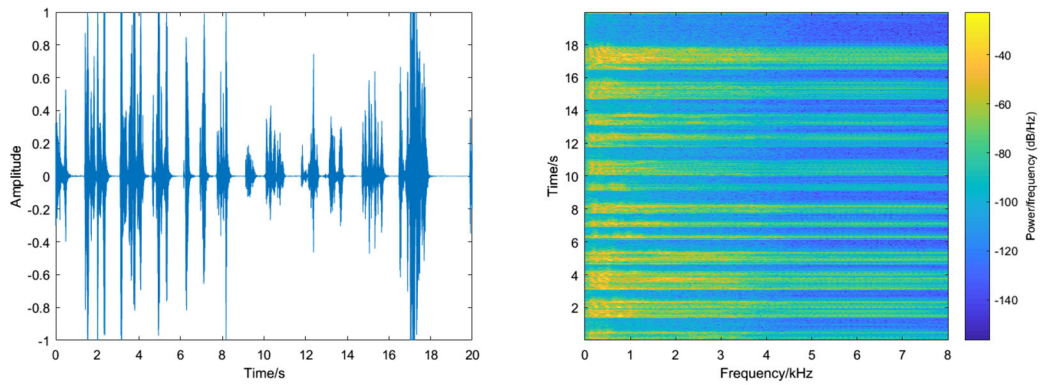
(a) No tampering



(b) 5% tampering

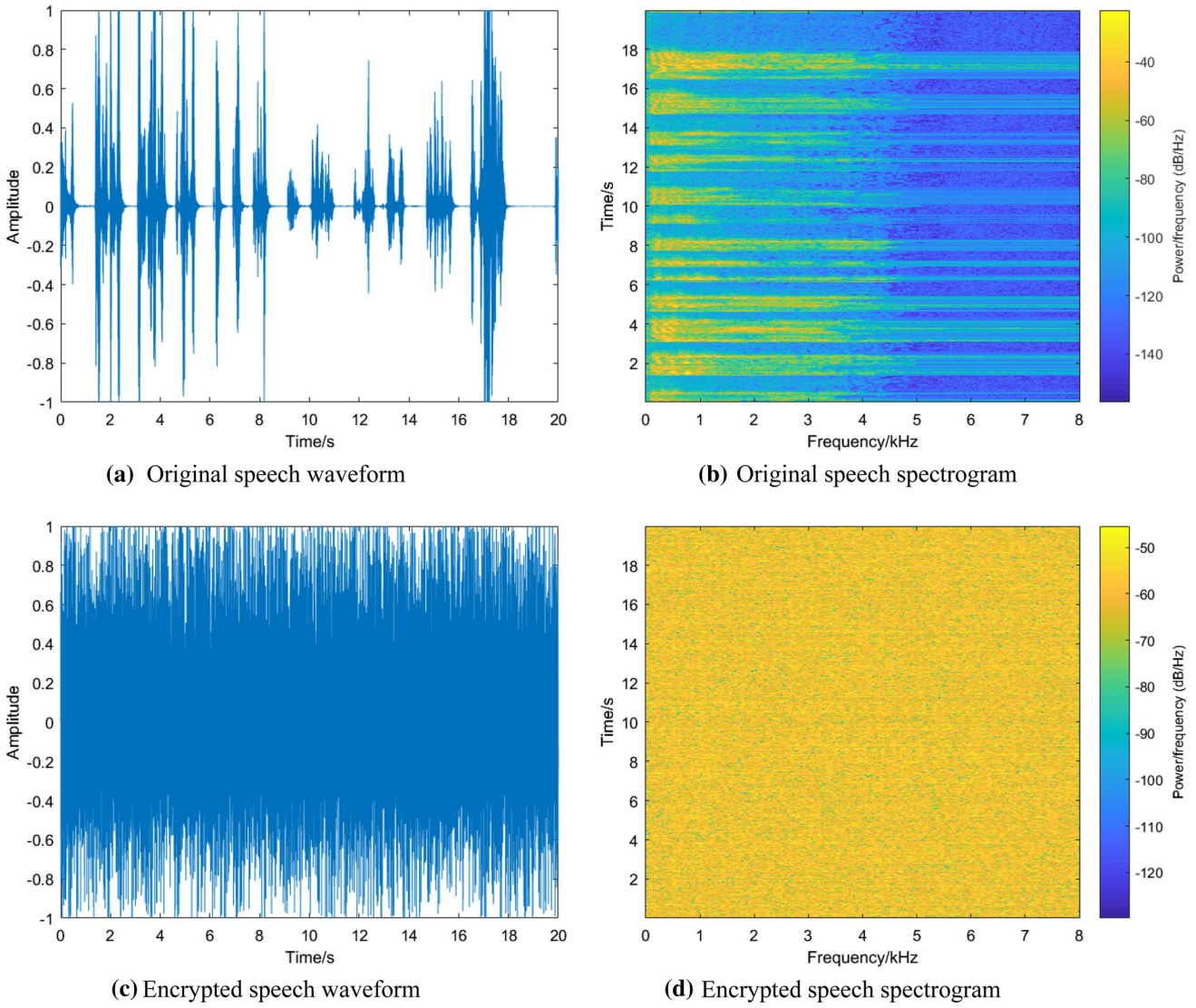


(c) 10% tampering



(d) 20% tampering

Fig. 6 The speech waveform and spectrogram after recovery under different tampering ratios



(a) Original speech waveform

(b) Original speech spectrogram

(c) Encrypted speech waveform

(d) Encrypted speech spectrogram

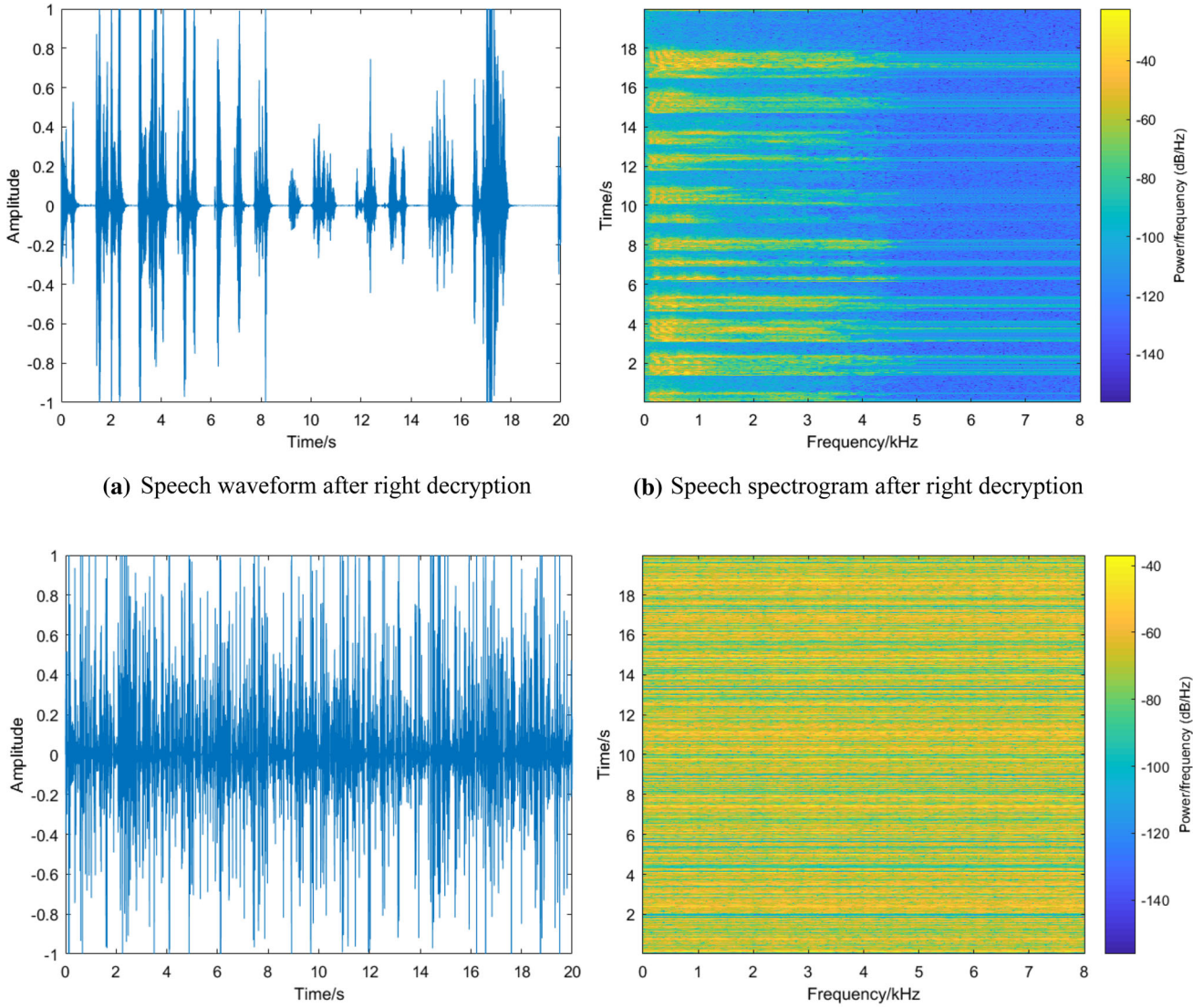
Fig. 7 Waveform and spectrogram of original speech and encrypted speech

In order to resist exhaustive attack, the encryption algorithm must have enough key space, and the key space that can resist the exhaustive attack requires more than 2^{100}. The initial value $K = (x_0,y_0,u_0,v_0)$ of the proposed method is double-precision floating-point data accurate to 12 decimal places, and the key space is $2 \times 10^{12} \times 2 \times 10^{12} \times 2 \times 10^{12} \times 2 \times 10^{12} = 16 \times 10^{48} \approx 2^{164}$, the key space is larger than 2^{100}, and if the parameters a_1, b_1, a_2, b_2 of the Henon sequences and the number of system iterations are taken into account, the encryption algorithm in this paper has a large enough key space to resist exhaustive attack and can meet the security requirements.

In order to verify the key sensitivity of the proposed scheme, this paper uses the correct key and the wrong key for decryption, the initial value of the wrong key is $K' = (x_0 + 10^{-12}, y_0 + 10^{-12}, u_0 + 10^{-12}, v_0 + 10^{-12})$. Figure 8 shows the waveform and spectrogram of the speech after right decryption and wrong decryption.

As shown in Fig. 8, if the key is correct, the encrypted speech can be decrypted, and even if the key changes a little, the entire chaotic sequence will change a lot, the encrypted speech cannot be decrypted correctly, and any useful information of the original speech cannot be obtained from the incorrectly decrypted speech. Therefore, the encryption algorithm in this paper has strong key sensitivity.

In order to further verify the encryption algorithm of this paper, this paper analyzes the PESQ and SNR of the encrypted speech and the decrypted speech, the experiment randomly selects 1000 speech for testing, and takes the average of the PESQ and SNR. Table 4 shows the compares the performance of encrypted speech and decrypted speech with the other method in Refs. [28, 29].



(a) Speech waveform after right decryption

(b) Speech spectrogram after right decryption

(c) Speech waveform after wrong decryption

(d) Speech spectrogram after wrong decryption

Fig. 8 Waveform and spectrogram of the speech after right decryption and wrong decryption

As shown in Table 4, compared with Ref. [28], the SNR of encrypted speech in this paper is better, and the recovery quality of decrypted speech is similar. Compared with Ref. [29], although the PESQ of encrypted speech in this paper is worse, it can still meet the need of security, and the recovery quality of decrypted speech in this paper is better. It can be seen from Table 4 that the PESQ and SNR of the encrypted speech are low, and the auditory quality of the encrypted speech is poor, it's difficult to obtain any information from the encrypted speech, and the PESQ is 4.5000 after decryption, the decryption effect is good, which has no impact on the audio quality of the speech. Therefore, the encryption method used in this paper can meet the security requirements.

5 Conclusions

In this paper, we have proposed an encrypted speech authentication and recovery scheme based on fragile watermarking, which constructs a fragile watermarking by converting the speech frame number into binary code, and embeds them into the encrypted speech by LSB. The watermark algorithm can judge the length of the original speech and construct the original watermark information through the extracted watermark to realize the tamper localization of sampling points. The encrypted speech that failed authentication is recovered by linear interpolation based on residuals, the recovery method recovers by the correlation between sampling points without additional information. Compared with the recovery method of linear interpolation, this paper uses the residual

error to optimize the result of linear interpolation to make it closer to the true value. Our experimental results show that the proposed scheme can realize the tampering detection of attacks such as mute, substitution, insertion, deletion, and can realize the tampering location of sampling points, and the tampering location error is small, which has basically no effect on the recovery of tampering, and can satisfy the actual needs of encrypted speech authentication and tamper recovery in the application field of speech secret communication or encrypted speech retrieval. Although this paper can realize the tampering positioning of sampling points in terms of authentication, it will generate errors of several sampling points, which will still have a certain impact on the case of many tampered positions, the future research work is to explore the smaller error of tampering positioning. In terms of tampering recovery, although tampering recovery can be achieved only through the correlation of sampling points, but when the tampering ratio is high, the recovery effect isn't as good as the tampering recovery scheme based on fragile watermarking, the next research work mainly focuses on the effect of tampering recovery, and realizes high-quality recovery of speech under high tampering ratio through deep learning and other methods.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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